

dynamically displays the result on the OLED.

Operating mode of the calculator

The calculator features three modes of operation: normal, scientific, and switching. Each is described below.

Normal. Used for basic arithmetic operations. In this mode, the keypad functions as follows:

- A - Multiplication (*)
- B - Addition (+)
- C - Subtraction (-)
- D - Division (/)
- * - Clear input
- # - Execute calculation

Scientific mode. Enables advanced operations. In this mode, the same keys are reassigned to:

- A - sin (value)
- B - cos (value)
- C - tan (value)
- D - sqrt (value)
- * - Clear input

Switching. The push button connected to pin D10 of the Arduino toggles between normal and scientific modes.

Software

The source code for the calculator has been developed using the Arduino IDE. The Adafruit SSD1306 OLED display library and the Keypad.h library are required. Once the libraries are added, the code can be uploaded to the Arduino by selecting the appropriate port and board. Fig. 3 shows a snippet of the source code.

Construction and testing

After uploading the source code to the Arduino Uno, the components should be assembled as shown in the circuit diagram (refer Fig. 2). A 12V, 2A adaptor must be connected to the Arduino Uno's DC power socket.

To operate the calculator, first check the current mode. If switching modes is required, press the push button (SW1). The selected mode will then be active and ready for use. **EFY**

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